Umm Al -Qura University
College of Computer



College of Computer Sciences

Project Title:
Reviving Colours

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Group: 3

Course Name :
Computer Graphics

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Members Task Table:

Member Name	Tasks Assigned
Student 1	Intro Scene + General Setup : -Designing Scene 1 Grayscale City, Story Text. -Setting up Project Structure (Files,Main functions, and Main loop). -Implementing Scene Navigation (Keyboard/Mouse). -Writing the User Manual .
Student 2	First Interactive Scene: -Designing Scene 2: Collecting the Paintbrush . -Implementing Main Character Movement. -Developing Collision Logic (Object touching) . -Applying Geometric Transformations (Translation + Rotation) for the character.
Student 3	Second Interactive Scene + Lighting: -Designing Scene 3: Gradual Color Restoration. -Applying Glow and Lighting effects. -Implementing Color Transition (Grayscale to Color conversion). -Applying Scaling transformations for elements upon appearance.
Student 4	Final Scene + Textures + Documentation: -Designing Scene 4: The Full Colored City. -Adding Textures (at least 2, such as Sky and Ground). -Compiling the full Project Report (Structure + Screenshots). -Preparing the Final Presentation.

Chapter 1: Introduction

Abstract: 1.1

Color Revival" is an interactive 2D project that " combines storytelling and gameplay. The goal is to restore life to a grayscale city by collecting items. This project demonstrates skills in transformations, color manipulation, and user interaction.

Introduction: 1.2

The project is set in a world without color. The player controls a hero through 4 scenes to find a magic brush and restore the city's vibrancy. It covers the technical implementation of animation, light effects, and smooth scene transitions.

Chapter 2: Methodology

Implementation Details : 2.1

In this project, we used [Mention Library OpenGL/GLUT]

Transformations: We applied Translation for character movement, Rotation for item animation, and Scaling for visual effects when items are collected.

Graphics: We implemented a Color Transition algorithm to shift the environment from grayscale to RGB.

Interaction: The project uses Keyboard input for movement and Mouse clicks for menu navigation.

Textures: Two main textures were applied to the sky and ground to enhance realism.

Chapter 3: Result

Screenshots : 3.1

Figure 1 : The Intro Scene in grayscale.

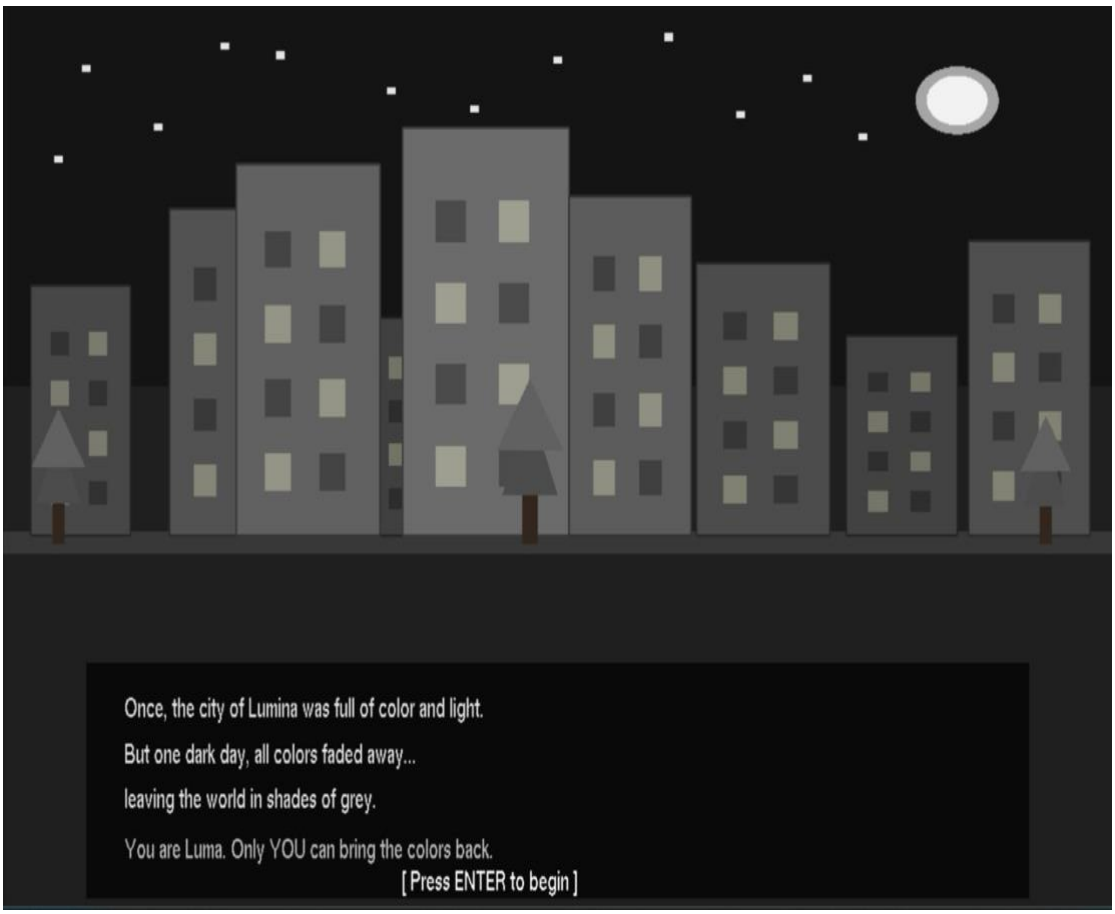


Figure 2 : The character collecting the magic brush.

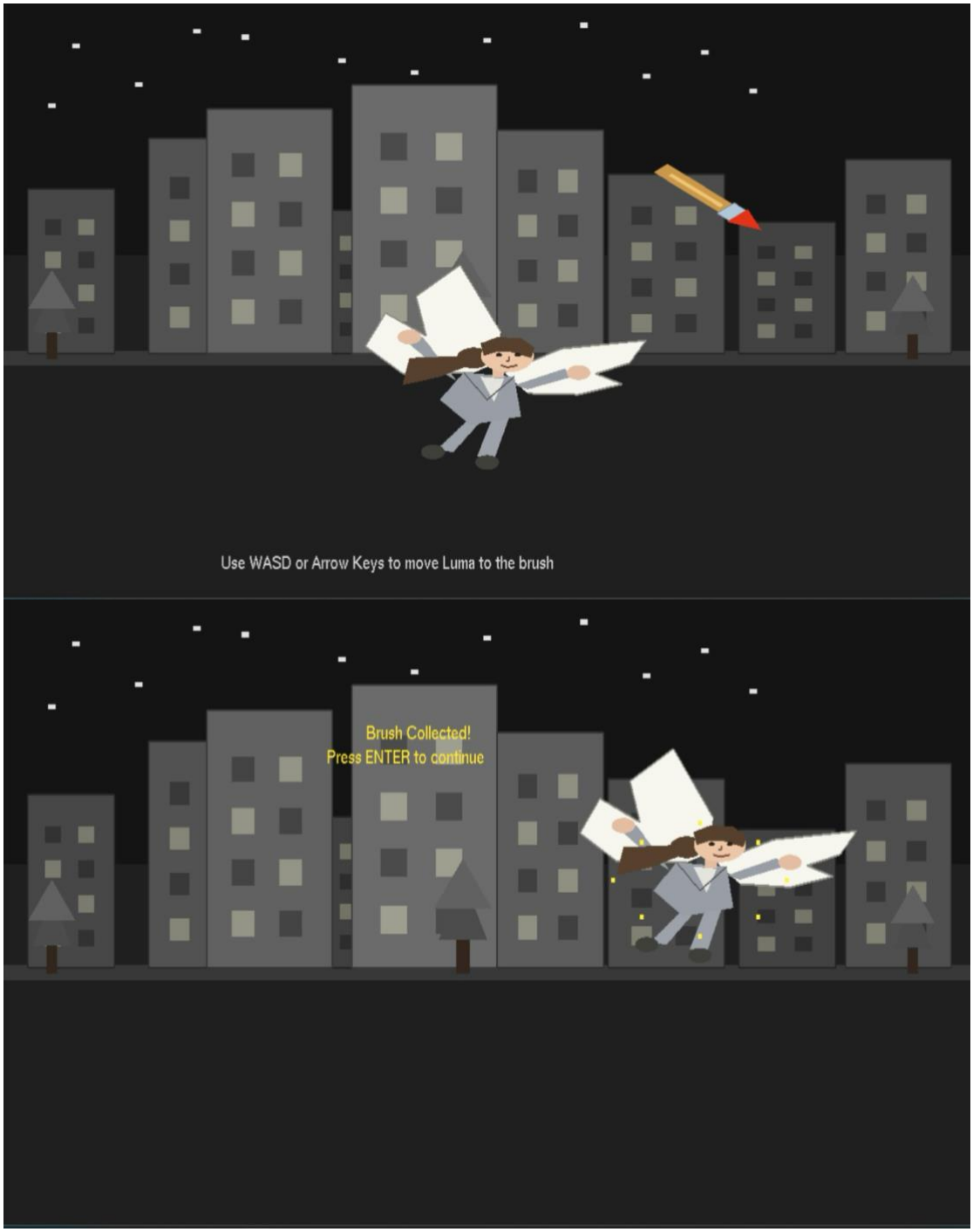
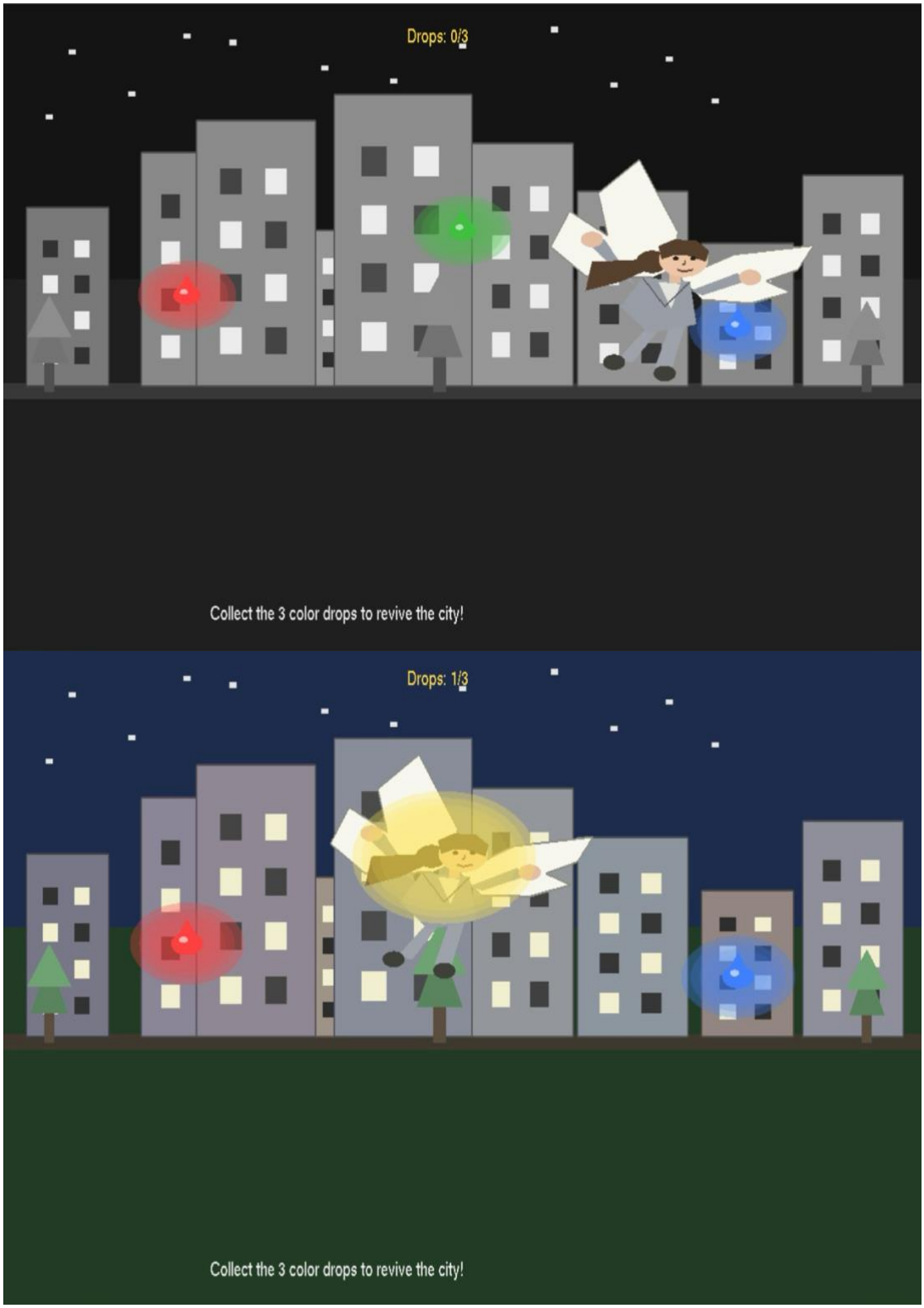


Figure 3: Colors gradually returning with glow effects.




Drops: 1/3
Collect the 3 color drops to revive the city!


Drops: 2/3
Collect the 3 color drops to revive the city!

Figure 4: The final fully colored city.



Conclusion & Future Work:

Conclusion:

We successfully implemented a four-scene interactive story that applies all the technical requirements of the course. Through this project, we learned how to effectively use transformations, textures, and collision logic to create an engaging visual experience. The team collaborated to ensure a seamless transition from grayscale to a fully .colored environment

Future Work:

In the future, this project could be expanded by

Transitioning the environment from 2D to 3D to add more depth.

Adding more complex levels and different types of visual puzzles.

Integrating background music that changes dynamically as colors are restored.

User Manual: Color Revival

How to Build :

Software Needed: Visual Studio

Libraries: OpenGL , GLUT, CMATH, STRING CSTDLIB libraries.

How to Run :

After building successfully, click the Run button

The game window titled "Color Revival" will appear.

Controls & Interaction:

Navigation: Press the Enter key to transition between scenes (e.g., from the Story scene to the Gameplay scene).

Movement: Use the Arrow Keys (Up, Down, Left, Right) to control the main character’s movement.

Interaction: Move the character toward the Paintbrush and other items to trigger the collision logic and collect them.

Exit: Press the Esc key to exit the application.

Goal: Collect all items in each scene to restore the city's colors and reach the final scene.

References:

Course Academic Materials:

-Nagoor, O. H. Computer Graphics Lecture Slides (CS2206). Umm Al-Qura,, 2026.

-Nagoor, O. H. Lab Tasks and Materials (CS2206).Umm Al-Qura, 2026.

-Nagoor, O. H. CG Project Instructions 2026 –

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Textures and Assets:

Sky Texture:

[https://pin.it/ZdDMmVWrg]: This texture was used to render the sky background in the final scene

Ground Texture:

[https://pin.it/7M1uWf7bH]: This texture was applied to the ground surfaces to enhance environmental realism